

SQL fundamentals

Exercise 2!

Aggregate Functions & Grouping

Q1 COUNT() Function

Write SQL query to find the total number of employees in the company.

```
SELECT COUNT (ID)
FROM Employees_db
```

Outcome

ID
10

Q2 SUM () function

Write SQL Script to find the total salary paid to all employees in the IT department

```
SELECT SUM (Salary) AS total_Salary_Paid,  
        department  
FROM    employees  
WHERE   department = 'IT'
```

Outcome

department	total_Salary
IT	220 000

Q3 AVG () function

Write SQL query to calculate the average salary of employees in the HR department.

```
SELECT AVG (salary)  
FROM emp
```

```
SELECT AVG (salary),  
       department  
FROM   employees;
```

Outcome

department	Salary
HR	48000

Q4 MIN() and MAX() function

Write SQL query to find highest and lowest salary in the company.

```
SELECT MIN(Salary) AS lowest-Salary-Paid  
       MAX(Salary) AS highest-Salary-Paid  
FROM   employees
```

Outcome

Highest-Salary	lowest-Salary
62000	48000

Q5 Group BY Statement

Write a SQL query to group employees by department and display the total salary-paid in each department.

```
SELECT department
       Sum(salary) as total_salary_paid
FROM   employees_db
GROUP BY department
```

Outcome

Department	total_salary_paid
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

Q6 Group BY and COUNT()

Write a SQL query to Count how many employees work in each city.

```
SELECT COUNT(ID) AS number-of-employees  
       City  
FROM   employees-db  
GROUP BY City
```

Outcome

City	Number-of-employees
New York	2
Los Angeles	2
Chicago	3
Houston	1
San Francisco	2

⑦ GROUP BY and ORDER BY

Write a SQL query to group employees by department, calculate the average salary in each department, and order the results by in descending order of average salary.

```
SELECT COUNT(*) AS number-of-employees,  
       department,  
       AVG(salary) AS Average-Salary  
FROM employees  
GROUP BY department  
ORDER BY Average-Salary DESC;
```

Outcome

Department	Average-Salary	DESC
IT	220 000	
Finance	119 000	
Marketing	105 000	
HR	99 000	

⑧ HAVING Clause

Write a SQL query to find departments where the total salary paid exceeds 100,000. (use GROUP BY and HAVING)

```
SELECT    Sum (Salary) AS total_Salary-Paid  
          department  
FROM      employees - db  
WHERE     Salary > 100,000
```

Outcome

Department	total_Salary-Paid
Marketing	105 000
Finance	119 000
IT	220 000

① Combine GROUP BY, HAVING and ORDER BY

Write a SQL query to list cities where more than one employee works, order by the number of employees in descending order.

```
SELECT COUNT(id) AS number of employees  
       City  
FROM employees - db  
WHERE City = > 1 employee(s)  
GROUP BY City  
ORDER BY employees DESC
```

employee id Department	City	Number of employees
	Chicago	3
	New York	2
	Los Angeles	2
	San Francisco	2

⑩ Combining Aggregate Functions

Write a SQL query to find the department with the highest average salary.

```
SELECT      Sum (salary) AS total_salary_paid,  
            department ,  
            AVG (salary) AS highest_average_salary  
FROM        employees_db ;
```

Outcome

Department
IT

Salary - AS - highest AVG
220 000